Explorations: Three Cross-Machine Findings

Zooming out from the machines and the meta report — ideas from the Collatz / generalized-Collatz literature, and what they turned up

0. What this is

A step back from the per-machine work to ask what the collection looks like as a whole, and to try directions we had not: the generalized-Collatz drift dichotomy (Matthews–Watts), the “missing backbone” our machine 1 report flagged as open, and the structural reason congruence invariants fail on the multiplicative machines. Five ideas were catalogued (IDEAS.md); all five — including the deepest, Baker-type perfect-powers — were pursued, yielding the six findings below. One is a **correction** to the meta report (now folded in); five are new positive structure — the mantissa backbone and its transfer operator, the no-congruence theorem, a self-similarity relation, the halting set computed backward, and the Baker / perfect-powers frontier. Everything is machine-verified.

1. The “divergent vs convergent cryptid” dichotomy is not where the report puts it

The meta report (§5) frames a dichotomy: machine 1 grows geometrically and is a “divergent” cryptid, machine 4 grow linearly and are “convergent,” and it implies the growth rate sets the *risk profile* (the $10^{-75,000}$ vs 10^{-11} residual-risk gap). Measuring the actual return-map multipliers shows this is misleading.

For each machine, take the first-return map to its section (F for machine 1, the reset maps for 2, 3, 4, the step map for Space Needle) and measure the mean log-ratio of consecutive return values — the Matthews–Watts drift:

machine	return drift (dlog)	return-time slope / index	per-step state growth	$\Sigma\ 1/C_n$
machine 1	0.87	≈ 0 (constant)	geometric	converges
machine 3	0.33	≈ 0 (constant)	geometric	converges
machine 4	0.99	0.84 (thinning)	linear	converges
Space Needle	0.65	— (map = step)	geometric	converges

Every return map is supercritical (drift 0.33–0.99, none critical), so the return values grow geometrically and the halting-risk sum $\Sigma\ 1/C_n$ converges for **all** machines, by the same mechanism. There is no divergent/convergent split in the halting risk. What actually differs is the **return frequency**: the return-time is constant in the index for machines 1 and 3 (frequent returns → geometric per-step growth) and grows exponentially in the index for machine 4 (thinning returns → linear per-step growth). This is a real structural difference — but it is about how fast the raw state grows per step, not about the halting mechanism.

Consequence. The residual-risk gap across the collection ($10^{-75,000}$ for machine 1 against far larger figures elsewhere) is a *verified-depth artifact*, not a qualitative dichotomy: machine 1’s orbit was pushed to $10^{150,000}$-scale return values while the others reached far smaller scales, so $1/C_n$ is smaller for machine 1 simply because C_n is larger. Pushed to equal depth, the risks would be comparable. All three machines — and the Space Needle — are the **same kind of cryptid for the halting question**: supercritical return map, geometrically spaced target, convergent risk sum. The meta report's §5 dichotomy should be reframed accordingly (see §4).

2. The mantissa "backbone" of machine 1, located

Machine 1's report proved F has no continuous 2-adic extension and showed the mantissa $m_k = \mathrm{frac}(\log_2 D_k)$ is *not* uniform ($\chi^2 \approx 162$), then flagged the true stationary density as the "missing ergodic backbone" — the concrete object a real proof would need. We compute it and identify its structure.

The dominant word gives $F(D) = 16D - 240\cdot 2^n + \ldots$ with $n = n_A(1, D)$, so on the leading scale

$$m \to m + \log_2(1 - 15\cdot 2^{n-\mathrm{floor}(\log 2\ D)}\cdot 2^{-m}) \pmod 1,$$

an explicit two-branch circle map whose branch is set by the exponent offset $n - \mathrm{floor}(\log_2 D)$. Measured over 40,000 cycles, that offset is –4 below a sharp breakpoint and –3 above it, and the breakpoint is exactly

$$m_* = \log_2(5/4) = 0.321928\ldots$$

— the mantissa at which 2^n jumps between D/16 and D/8. The two branches have opposite character: below m_* the map spreads mass across the whole circle (mixing), above it the map contracts sharply toward $m \approx 0.5$. The resulting stationary density is **elevated (up to 1.3×) on $[0, \log_2(5/4))$ and depressed (down to 0.85×) on $[\log_2(5/4), 1)$**, with the transition exactly at the breakpoint.

This identifies the object the report called missing: the mantissa backbone is the invariant measure of an explicit piecewise circle map with breakpoint $\log_2(5/4)$. It is (like most such measures) probably not elementary in closed form, but it is now a concrete, localized target rather than an unknown — and the elevated-below / depressed-above shape is the first structural fact established about it. (This is also why the naive uniform-equidistribution assumption failed: the map is genuinely non-uniform, contracting on the larger of its two branches.)

The transfer operator confirms and quantifies it. Binning the mantissa into 60 cells and forming the empirical Markov (Perron–Frobenius) operator, its stationary distribution matches the direct histogram to $L_1 = 0.0001$ — the mantissa process is a genuinely mixing Markov chain on the circle whose invariant measure is the density — with a **spectral gap of 0.77** ($|\lambda_2| = 0.23$), so the backbone is a strongly attracting measure, not a slow drift. The stationary density is 1.12× uniform below $\log_2(5/4)$ and 0.94× above, with a **jump of about 0.2 exactly at the breakpoint** (the density steps from ≈1.22 down to ≈1.01 across the two adjacent cells). The next step is a fine-grid Perron–Frobenius solve of the analytic two-branch map to get the density to arbitrary precision.

3. Why no congruence can decide the multiplicative machines

For machine 3 and the Space Needle the halting event is multiplicative — the orbit must hit an exact power of q (q = 3, resp. 2). Our searches for a separating modulus came up empty; here is the structural reason they had to, promoting the empirical result to an impossibility.

Both maps branch on the **q-adic valuation** of the state: machine 3 divides by 3 exactly $v_3(a)$ times; the Space Needle's step reads $v_2(b)$. But for any modulus M coprime to q, the valuation $v_q(x)$ is **independent of x mod M** (by the Chinese Remainder Theorem, every residue class mod M realizes every valuation — verified: for M = 3, 5, 7, 9, 15 every class mod M attains a full range of 2-adic valuations). So the map is not a function of x mod M at all: no congruence closure mod M even exists, let alone one that separates the orbit from the target.

This is the **Space Needle / machine 3 analogue of the Hydra family q-adic branch-memory theorem**: on the walk-absorption machines the value's residues carry only bounded branch history; on the multiplicative machines the halting-relevant data (the valuation) lives on the q-adic side, orthogonal to residues mod any M coprime to q. In both families the reason congruence deciders fail is the same — the arithmetic that decides halting is orthogonal to the arithmetic a congruence can see — and it is now a theorem for three of the collection's types, not a search outcome.

4. A self-similarity relation for the Space Needle

The Space Needle's map $b \to b + v_2(b) + (3/2)(\mathrm{oddpart}(b) - 1)$ obeys an exact **scaling relation**: doubling the argument shifts the image by a controlled amount,

$$step(2b) = step(b) + b + 1 \quad \text{(for every non-halting b)},$$

verified with no exceptions for b < 200,000. (It follows directly: $v_2(2b) = v_2(b) + 1$ while the odd part is unchanged.) This is a genuine self-similarity — the map is not scale-invariant, but its failure to be is the explicit affine cocycle b + 1. The analogue on machine 3 is its divide-chain lemma ($b \to b + (N - M) + j$ collapses a factor 3^j in one step): both multiplicative machines carry an exact renormalization of their q-scaling. It did not (yet) yield a decision, but it is the kind of exact structure a renormalization argument would need, and it cleanly explains why the halt targets at different scales are linked (2b is a power of 2 iff b is).

5. The Space Needle halting set, computed from the target backward

Turning the halting question around: instead of following the orbit forward, enumerate the **halting set** — every b whose orbit reaches a power of 2 — by breadth-first search backward from the powers of 2 through the inverse map (the reverse-decider idea from bbchallenge). Each target has O(1) preimages (one odd-branch, a few even-branch), so the set is thin and exactly computable.

Below 2,000,000 there are only **16 halting seeds** {7, 103, 312, 352, 372, 1639, …}, density 8×10^{-6}, and **the start b = 6 is provably not among them**. The density falls geometrically by dyadic scale — about 1.6×10^{-2} at 2^6 down to $\approx 10^{-6}$ at $2^{18–20}$ — with roughly one seed per octave, so the halting set is only about **logarithmically many below N**. This corroborates non-halting from the target side, independently of the forward orbit, and makes the sparsity exact: a forward orbit visiting ~log N values below N against a target of ~log N values in a space of N collides with probability $\sim(\log N)^2/N \to 0$ —

the Borel–Cantelli estimate of Finding 1, now with a rigorously enumerated target rather than a heuristic density. (The same enumeration applies verbatim to machine 3 with powers of 3.)

6. The perfect-powers question: Baker reaches the runs, not the orbit

The deepest idea was to attack the multiplicative machines' halting (does the orbit hit q^k?) with the tools that excluded Collatz cycles — Baker's linear forms in logarithms and the Bilu–Hanrot–Voutier primitive-divisor theorem. These apply to sequences with S-unit (smooth) structure. The first thing to check is whether the orbit has it, and it does not:

The orbit is not an S-unit sequence. The Space Needle values 6, 10, 17, 41, 101, 251, … are smooth only twice before $b_2 = 17$ and $b_3 = 41$ bring in large primes; the factorizations are generic thereafter. So Baker cannot be pointed at the orbit sequence directly — the first-order obstruction, and the reason a decision is not simply a matter of citing a theorem.

But the geometric runs do have the structure, and Baker — via its elementary p-adic core — bites there. During an odd run the map is exactly affine: $b_n - 1 = (5/2)^n(b_0 - 1)$, so a halt $b_n = 2^k$ is the three-term S-unit equation $2^{n+k} - 2^n = 5^n(b_0 - 1)$. Lifting-the-exponent gives $v_5(2^k - 1) = 1 + v_5(k/4)$ for $4 \mid k$, so $5^n \mid 2^k - 1$ forces $k \geq 4 \cdot 5^{n-1}$. The unconditional consequence:

$$\textit{at orbit scale } B\textit{, a halting run has length at most} \approx 1 + \log_5(\log_2 B),$$

a log–log bound: at scale 2^{20} a halt cannot follow a run longer than 2; at $2^{150{,}514}$ (machine 1's verified horizon), longer than 7; even at $2^{10\hat{}9}$, longer than 13. So **a halt can never occur in the interior of a geometric ascent** — it must land on the power essentially at a run boundary. Empirically the odd runs reach length 16, but a length-3 halting run would already require the orbit to be past 2^{97}. This is, as far as we know, the first time linear-forms-in-logarithms has been aimed at a cryptid's *halting* family (the meta report notes it had been aimed only at Collatz cycles).

Where it stops. The bound is per-run; the branch word *between* run boundaries is orbit-determined and grows without bound (digit consumption), so per-run finiteness does not aggregate into a global bound. That is exactly the Collatz obstruction, now located precisely: Baker decides each fixed word, and the undecidability lives entirely in the unboundedness of the word. A full decision would need to control the word sequence itself — which is the open problem. What we have is a genuine, unconditional partial result (no interior-of-run halts) and an exact map of the frontier.

7. What changed, and what is left

Done. The Finding 1 correction has been folded into the meta report: §5 is reframed (the divergent/convergent labels describe per-step growth — frequent vs thinning returns — not two halting-risk regimes; all machines share the supercritical-return, convergent-Σ1/C mechanism, with the residual-risk numbers as depth artifacts), P8 is stated as the *right* dichotomy with all four machines on its convergent side, and the status labels now read “geometric / linear growth” rather than “divergent / convergent cryptid.”

All five catalogued ideas are now tried (Findings 1–6): the drift dichotomy, the mantissa backbone and its transfer operator, the no-congruence theorem, self-similarity, backward reachability, and the perfect-powers / Baker direction. Together they turned three of the collection's open flags into structure — the backbone is located and measured, the congruence and Baker obstructions are pinned exactly, the halting set is enumerated — and produced one unconditional partial result (no interior-of-run halts).

What is left is the open problem itself. Every tool now has a sharp reason it stops at the same place: the branch word is orbit-determined and grows without bound. Congruences cannot read it (Finding 3), Baker decides each fixed word but not their unbounded sequence (Finding 6), and the mantissa/valuation process that generates it is mixing with a positive spectral gap (Finding 2) — structured enough to model precisely, random enough to defeat every finite invariant. A decision would require controlling that word sequence, which is exactly what makes these machines cryptids. The honest end state is not a decision but a complete map of why one is out of reach, with the frontier drawn tool by tool.

Artifacts (collatz/explorations/): IDEAS.md, growth_law.py, dichotomy.py, mantissa.py, mantissa_map.py, transfer_operator.py, selfsimilar.py, backward.py, baker.py. All findings machine-verified. The Finding 1 correction is folded into the meta report (§6).